



Mainak Saha

27th December, 1997
Agartala, Tripura (W), India

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in https://linkedin.com/in/mainak-saha/

EDUCATION

- **National Institute of Technology Agartala** 2024 – present
Pursuing Doctor of Philosophy (PhD) in Computer Science and Engineering (CSE)
- **Tripura University (Central University)** 2022
Completed Master of Technology (M.Tech) in Computer Science and Engineering (CSE)
- **Godavari Institute of Engineering and Technology (affiliated to JNTUK)** 2020
Completed Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE)
- **Assam Rifles Public School** 2016
Completed Higher Secondary Certificate (HSC)
- **Sri Krishna Mission School** 2014
Completed Secondary School Certificate (SSC)

WORK EXPERIENCE

- **Techno College of Engineering Agartala** 06-08-2025 – Present
Assistant Professor, Department of Computer Science and Engineering (CSE)
Agartala, Tripura, India
- **ICFAI University Tripura** 29-07-2024 – 05-08-2025
Visiting Professor, Department of Computer Science and Engineering (CSE)
Kamalghat, Tripura, India
- **SKARMA SOFTTECH PRIVATE LIMITED** 08-07-2024 – 10-07-2025
Software Developer
Agartala, Tripura, India
- **K L University** 04-09-2023 – 06-07-2024
Assistant Professor, Department of Computer Science and Engineering (CSE)
Hyderabad, Telangana, India
- **K J Institute of Engineering and Technology** 01-11-2022 – 31-08-2023
Assistant Professor, Department of Computer Science and Engineering (CSE)
Vadodara, Gujarat, India

TRAINING/INTERNSHIP

- **Airport Authority of India (AAI)** 17-12-2018 – 30-12-2018
Intern
– Gained exposure to various air navigation and communication systems
Agartala, Tripura, India
- **Bharat Sanchar Nigam Limited (BSNL)** 31-12-2018 – 12-01-2019
Intern
– Learned the fundamentals of networking and its practical applications
Agartala, Tripura, India
- **North Eastern Electric Power Corporation Limited (NEEPCO)** 13-05-2019 – 28-05-2019
Intern
– Worked on IT solutions for power generation and monitoring systems
Laxmipur, Tripura, India
- **National Informatics Centre (NIC)** 29-05-2019 – 15-06-2019
Intern
– Developed practical skills in web development through hands-on training
Agartala, Tripura, India

LANGUAGES

- English
- Hindi
- Bengali (mother tongue)
- Telugu
- Gujarati

RESEARCH INTEREST

- Artificial Intelligence
- Machine Learning
- Computer Vision
- Image Processing
- Pattern Recognition
- Deep Learning

PROJECTS

- **Deep Fake Object Generation using Classifier Posterior GAN (CP-GAN)** 2021-2022
Tools & technologies used: Deep Learning, Pytorch, GAN
- **Detection of Malaria in Blood Smear Using Image Processing and Machine Learning** 2019-2020
Tools & technologies used: Image Processing, Machine Learning

PUBLICATIONS

Journal(s)

- **Mainak Saha**, M. Trinath Basu, Arpita Gupta, K. Ashrith, Chevella Vamshi Vardhan Reddy, Shashanth Reddy, and Rohith Reddy, "*Enhancing Credit Card Security Using RSA Encryption and Tokenization: A Multi-Module Approach*," *International Journal of Informatics and Communication Technology (IJ-ICT)*, vol. 14, no. 1, pp. 132–140, 2025. DOI: 10.11591/ijict.v14i1.pp132-140
- Miriyala Trinath Basu, **Mainak Saha**, Arpita Gupta, Sumit Hazra, Shahin Fatima, Chundakath House Sumalakshmi, Nallagopu Shanvi, Nyalapatla Anush Reddy, Nallamalli Venkat Abhinav, and Koganti Hemanth, "*UniMSE: A Unified Approach for Multimodal Sentiment Analysis Leveraging the CMU-MOSI Dataset*," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 39, no. 3, pp. 2032–2042, 2025. DOI: 10.11591/ijeecs.v39.i3.pp2032-2042

Conference Proceeding(s)

- **Mainak Saha**, Sourav Dey Roy, Santanu Das, and Mrinal Kanti Bhowmik, "*A Comprehensive Survey on Deep Fake Object Creation and Detection: Challenges, and Future Directions*," in *Advances in Computational Intelligence and Its Applications*, CRC Press, 2024, pp. 105–112. DOI: 10.1201/9781003488682-15 (Best Paper Award)
- **Mainak Saha**, Rabin Bhaumik, Abadhan Ranganath, Miriyala Trinath Basu, and Pavan Kumar Pagadala, "*A Comprehensive Survey of Deep Learning Techniques in Malaria Diagnosis*," in *Proceedings of the 2024 IEEE 6th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*, IEEE, 2024, pp. 61–66. DOI: 10.1109/ICCCMLA63077.2024.10871574
- **Mainak Saha**, Pavan Kumar Pagadala, and Miriyala Trinath Basu, "*Cyber Deceptions: Unveiling the Threat of Deepfakes in Digital Realms*," in *Proceedings of the 2024 IEEE 6th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*, IEEE, 2024, pp. 113–116. DOI: 10.1109/ICCCMLA63077.2024.10871909
- Sushanta Das, **Mainak Saha**, Ranjita Das, and Tannistha Pal, "*Underwater Object Detection: Synergizing Image Processing Techniques with Deep Learning Frameworks*," accepted in the *Second International Conference on Advanced Multidisciplinary Research and Innovation Technologies (AMRIT 2024)*
- Tania Bhattacharya, Yagnyasene Sengupta, **Mainak Saha**, Sanghita Deb, Mainak Dhar, Jani Chakraborty, and Pavan Kumar Pagadala, "*A Comprehensive Study on Smart Irrigation Systems Using IoT and Machine Learning for Enhanced Water Efficiency and Agricultural Productivity*," accepted in the *6th International Conference on Recent Trends in Machine Learning, IoT, Smart Cities and Applications (ICMISC 2025)*.
- Sanghita Deb, Yagnyasene Sengupta, **Mainak Saha**, Tania Bhattacharya, Debajyoti Debnath, SK Shamim Aktar, and Pavan Kumar Pagadala, "*A Comprehensive Study on IoT-Enabled Smart Home Technology for Elderly Health Care and Safety*," accepted in the *6th International Conference on Recent Trends in Machine Learning, IoT, Smart Cities and Applications (ICMISC 2025)*.

- **Mainak Saha**, Sushanta Das, Tannistha Pal, and Prakash Duraisamy, "*Vision Transformer-Based Framework for Lung Cancer Detection in CT Scans*," accepted in the *16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT 2025)*.
- Sushanta Das, **Mainak Saha**, Tannistha Pal, and Prakash Duraisamy, "*Automated Highlighting of New Clinical Information in EHR Notes Using SMOTE and PCA-Based Classification*," accepted in the *16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT 2025)*.
- Deeptanu Choudhury, **Mainak Saha**, Tania Bhattacharya, and Sanghita Deb, "*A Novel Multimodal Deep Learning Framework for Music Genre Classification: Integrating Audio Features and BERT-Enhanced Lyrical Analysis*," accepted in the *3rd International Conference on Recent Advances in Artificial Intelligence & Smart Applications (RAAISA 2025)*, 14–15 December 2025.
- **Mainak Saha**, Deeptanu Choudhury, and Barnali Chowdhury, "*QVSN: A Hybrid Quantum-Enhanced AI Framework for Efficient CCTV Video Footage Summarization*," accepted in the *4th International Conference on Advanced Communications and Machine Intelligence (MICA 2025)*, 21–24 December 2025, The ICFAI University, Tripura, India.
- Deeptanu Choudhury, **Mainak Saha**, Tania Bhattacharya, and Sanghita Deb, "*A Multimodal Approach for Automatic Music Genre Classification Using Audio and Lyrical Features*," accepted in the *4th International Conference on Advanced Communications and Machine Intelligence (MICA 2025)*, 21–24 December 2025, The ICFAI University, Tripura, India.

Book Chapter(s)

- **Mainak Saha**, Rabin Bhaumik, Shriya Chakraborty. 2026. *Harnessing AI for Waste Management and Circular Economy: Enhancing Recycling and Resource Efficiency.*, 71–87. doi:10.1108/978-1-83662-996-220251005.
- **Mainak Saha**, Sushanta Das. 2026. *Optimized Plant Identification Through Leaf Feature Extraction Using Deep Learning and Advanced Digital Image Processing Techniques.*, 89–104. doi:10.1108/978-1-83662-996-220251006.
- **Mainak Saha**, Rabin Bhaumik and Tannistha Pal, "*Comparative Analysis of Automated Malaria Cell Classification: EfficientNet-B0 Transfer Learning Versus Traditional Machine Learning*," in *Artificial Intelligence for Predictive Healthcare: Towards Personalized Treatment and Disease Prevention*, CRC Press, pp. 138–151.

Patent

- Parijata Majumdar, Sushmita Paul, Deeptanu Choudhury, Ankit Kishore Sen, **Mainak Saha**, M. Trinath Basu, "*Learning Disability Detecting AI Device*", India, Patent No. 438185-001, 13/01/2025. **(Published and Granted)**

Authored Book

- Dr. Parijata Majumdar, Dr. Sanjoy Mitra, **Mainak Saha**, Biswaraj Roy, "*Machine Learning*", Aryan Publishing House, 2025, ISBN 9788198761934. **(Published)**

PAPER PRESENTATION

- **Presented Paper:** *A Comprehensive Survey on Deep Fake Object Creation and Detection: Challenges, and Future Directions*
Conference: International Conference on Advances in Computational Intelligence and its Applications (ICACIA-2023)
Organized by: K L University, Hyderabad **Date:** 23-02-2023 to 25-02-2023 **Place:** Hyderabad, India
- **Presented Paper:** *Underwater Object Detection: Synergizing Image Processing Techniques with Deep Learning Frameworks*
Conference: Second International Conference on Advanced Computing, Machine Learning, Robotics and Internet Technologies (AMRIT-2024)
Organized by: NIT Agartala and Assam University **Date:** 08-11-2024 to 09-11-2024 **Place:** Agartala, India

- **Presented Paper:** *Cyber Deceptions: Unveiling the Threat of Deepfakes in Digital Realms*
Conference: 6th IEEE International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCMLA 2024)
Organized by: CMR Engineering College, Hyderabad **Date:** 19-10-2024 to 20-10-2024 **Place:** Hyderabad, India
- **Presented Paper:** *A Comprehensive Survey of Deep Learning Techniques in Malaria Diagnosis*
Conference: 6th IEEE International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCMLA 2024)
Organized by: CMR Engineering College, Hyderabad **Date:** 19-10-2024 to 20-10-2024 **Place:** Hyderabad, India
- **Presented Paper:** *Vision Transformer-Based Framework for Lung Cancer Detection in CT Scans*
Conference: 16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT 2025)
Organized by: IIT Indore, Madhya Pradesh, India **Date:** 06-07-2025 to 11-07-2025 **Place:** Indore, India

TECHNICAL SKILLS

- **Operating Systems:** Windows XP, 7, 8, 8.1, 10,11 and UBUNTU
- **Languages:** C, C++, Core Java, Python
- **Web Technologies:** HTML, CSS, JavaScript
- **Databases:** Oracle SQL, MySQL
- **IDEs:** Code::Blocks, Eclipse, Microsoft Visual Studio, Spyder, Jupyter notebook
- **Text Editors:** Notepad++, Sublime Text, Visual Studio Code
- **Typography:** LaTeX, Overleaf, Beamer, Microsoft Office
- **CMS:** WordPress

AWARDS/ACHIEVEMENTS/CERTIFICATION

- **Endorsed Faculty Member for Degree Engineering** 06-05-2023
Recognized by Gujarat Technological University (GTU)
- **Best Paper Award in ICACIA-2023** 25-02-2023
Organized by Department of CSE, KL University, Hyderabad
- **AWS Academy Graduate: Cloud Architecting** 21-10-2023
Completed a 40-hour course offered by AWS Academy
- **Certifications in C Programming, Python Core, HTML, and SQL** 06-12-2018
Awarded by SoloLearn (<https://www.sololearn.com>)

WORKSHOP/FDP/WEBINAR ORGANIZED

- **Organizing Committee, Impactech 2K24 Hackathon** 16-03-2024 to 17-03-2024
Organized by Department of CSE, KL University, Hyderabad Hyderabad, India
- **Organizing Committee, 5 days EPAT24 Workshop on latest AI Technologies** 28-05-2024 to 01-06-2024
Organized by Department of CSE, KL University, Hyderabad Hyderabad, India
- **Coordinator, Code & AI Summer Internship 2024 on AI and ML** 20-05-2024 to 29-06-2024
Organized by Department of CSE, KL University, Hyderabad Hyderabad, India
- **Organizing Committee, 5 days FDP on Prompt Engineering and Generative AI** 04-06-2024 to 08-06-2024
Organized by Department of CSE, KL University, Hyderabad Hyderabad, India
- **Organizing Committee, 1 Day Workshop on LATEX** 22-08-2025
Organized by Department of CSE, Techno College Of Engineering Agartala Agartala, India
- **Organizing Committee, 1 Day Workshop on Industry 4.0** 31-10-2025
Organized by Department of CSE, TCEA in collaboration with CTTC Bhubaneswar Agartala, India
- **Organizing Committee, Seminar on AI & Robotics and the Future of Engineering** 06-02-2026
Organized by Department of CSE & ME, TCEA in collaboration with IEI Tripura Agartala, India

WORKSHOP/FDP/WEBINAR ATTENDED

- **Winter School on Deep Learning: From Perceptrons to Transformers** 21-01-2022 to 12-03-2022
Organized by Indian Statistical Institute (ISI), Kolkata Kolkata, India
- **2 Days Workshop on Unsupervised Learning** 26-03-2022 to 27-03-2022
Organized by Department of Computer Science and Engineering, Tripura University Agartala, India
- **1 Day Workshop on Interpreting Machine Learning Model for Image Understanding** 09-07-2022
Organized by Department of Computer Science and Engineering, Tripura University Agartala, India
- **6 Days AICTE-ATAL FDP on Telemedicine 6.0** 11-16 December 2023
Organized by BHARAT INSTITUTE OF ENGINEERING AND TECHNOLOGY, Hyderabad Hyderabad, India
- **1 Day State-Level Capacity Building Workshop on AI for Good Governance** 16-05-2025
Organized by Directorate of Information Technology, Government of Tripura, Agartala Agartala, India
- **NEP 2020 Orientation and Sensitization Programme on “Research and Development”** 09–19 June 2025
Organized by UGC-MMTTC, Assam University, Silchar in collaboration with ICFAI University Tripura Online
- **Short Term Programme on Blockchain and Cybersecurity for Cyber Threat Prevention** 07-12 July 2025
Organized by the UGC Malaviya Mission Teacher Training Centre (MMTTC), IIT (ISM) Dhanbad Dhanbad, India
- **GIAN Course on Biological Cell Imaging for Diagnosis and Prediction of Cancer** 27-31 October 2025
Organized by Department of Computer Science and Engineering, Tripura University Tripura, India
- **6 Days AICTE-ATAL FDP on Deep Learning for Computer Vision** 03-11-2025 to 08-11-2025
Organized by Tripura Institute Of Technology, Agartala, Tripura Online
- **5 Days International FDP on “Impact and Influence of AI”** 02-02-2026 to 06-02-2026
Organized by Department of Computer Science, Sister Nivedita University, India Online
- **1 Day National Workshop on “Artificial Intelligence for Educational Institutions”** 10-02-2026
Organized by National Institute of Technology Jamshedpur Jamshedpur, India
- **5 Days Lecture Session and Ideathon on “AI Focused Pre-Summit Event”** 09-02-2026 to 13-02-2026
Organized by Department of Information Technology, Tripura University Agartala, India
- **5-Day FDP on “The Adaptive Educator: Theory-to-Application”** 18-02-2026 to 24-02-2026
Organized by Institute of Engineering and Management & HRDC, IEM-UEM Group Kolkata, India
- **Bootcamp on Applied Data Science with Deep Learning** 23-02-2026 to 27-02-2026
Organized by C-DAC Noida & FutureSkills Prime (MeitY & NASSCOM) Noida, India
- **1 Day Webinar on “Green AI in Healthcare”** 27-02-2026
Organized by OmDayal Group of Institutions in association with IEEE Computer Society Kolkata Section Online
- **6 Days National Level FDP on “AI for Agricultural Research”** 02-03-2026 to 07-03-2026
Organized by PCCOE Pune in association with IEEE Computational Intelligence Society Pune, India

PROFESSIONAL MEMBERSHIPS

- Lifetime Member, International Association of Engineers (IAENG), Member ID: 378453, Since 2024.
- Member, Internet Society (ISOC), Member ID: 2339350, Since 2025.

EXTRA-CURRICULAR ACTIVITIES

- Coordinated an industrial visit to *Bhaskaracharya Institute for Space Applications and Geo-Informatics*.
- Served as External Examiner for project viva at *Sigma Institute of Engineering*, Vadodara, Gujarat.
- Delivered a guest lecture on Artificial Intelligence at *SD Memorial HS School*, Jampuijala, Tripura.
- Coordinated a social outreach program “*Sandhyaneer – A Visit to an Old Age Home*” on Engineer’s Day.
- Served as External Examiner for project viva at *NIELIT Agartala*, Laxmipur, Tripura.
- Coordinator for *Prayukti 2026 Techfest* at Techno College of Engineering Agartala (25–28 Feb 2026).

DECLARATION

I do hereby declare that all the above information given by me are true to the best of my knowledge and belief.

Place: Agartala

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